# A. Find Minimum Operations

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: bitmasks, brute force, greedy, math, number theory |
https://codeforces.com/problemset/problem/2020/A

You are given two integers $n$ and $k$.

In one operation, you can subtract any power of $k$ from $n$. Formally, in one operation, you can replace $n$ by $(n-k^x)$ for any non-negative integer $x$.

Find the minimum number of operations required to make $n$ equal to 0.

## Input

Each test contains multiple test cases. The first line contains the number of test cases $t$ ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains two integers $n$ and $k$ ($1 \le n, k \le 10^9$).

## Output

For each test case, output the minimum number of operations on a new line.

## Example

| Input |
| --- |
| 6<br>5 2<br>3 5<br>16 4<br>100 3<br>6492 10<br>10 1 |

| Output |
| --- |
| 2<br>3<br>1<br>4<br>21<br>10 |

## Note

In the first test case, $n = 5$ and $k = 2$. We can perform the following sequence of operations:

1. Subtract $2^0 = 1$ from 5. The current value of $n$ becomes 5 - 1 = 4.
2. Subtract $2^2 = 4$ from 4. The current value of $n$ becomes 4 - 4 = 0.

It can be shown that there is no way to make $n$ equal to 0 in less than 2 operations. Thus, 2 is the answer.

In the second test case, $n = 3$ and $k = 5$. We can perform the following sequence of operations:

1. Subtract $5^0 = 1$ from 3. The current value of $n$ becomes 3 - 1 = 2.
2. Subtract $5^0 = 1$ from 2. The current value of $n$ becomes 2 - 1 = 1.
3. Subtract $5^0 = 1$ from 1. The current value of $n$ becomes 1 - 1 = 0.

It can be shown that there is no way to make $n$ equal to 0 in less than 3 operations. Thus, 3 is the answer.